Master in Information Technology (MIT) – Professional Track
OBE Framework and Implementation

PROGRAM OBJECTIVES

- Participating in professional development or postgraduate study in order to collaborate in diverse team environments and work effectively in the IT field to make a positive contribution to society through leadership positions and/or pursuing a research career.
- Integrate and apply IT-based solutions effectively in their practice to solve problems and meet the desired needs of society.
- Demonstrate a thorough understanding of the broad concepts of information technology while exercising professionalism and ethical judgment in the development, implementation, and management of information technology systems.

PROGRAM OUTCOMES

Graduates of the Master of Information Technology program will possess advanced knowledge and practical expertise in the field of information technology, equipping them for leadership roles in diverse professional settings. The MIT Program at Partido State University aligns with **Level 7 of the Philippine Qualifications Framework (PQF)** as mandated in **CMO No. 15, s. 2019**, which specifies that graduate programs must demonstrate:

ADVANCED KNOWLEDGE AND SKILLS IN INFORMATION TECHNOLOGY

PO1. Apply advanced IT concepts, frameworks, and methodologies to design innovative, secure, and scalable technological solutions that address complex organizational and societal problems.

PO2. Demonstrate mastery of specialized IT domains—including systems design, data analytics, cybersecurity, and IT management—through interdisciplinary and industry-aligned approaches.

SELF-DIRECTED RESEARCH

PO3. Independently conceptualize, design, and conduct high-quality IT research or capstone projects that contribute to innovation, digital transformation, and knowledge creation.

PO4. Critically evaluate existing IT literature, technologies, and practices to propose improvements, emerging trends, or novel approaches.

LIFELONG LEARNING WITH A HIGH DEGREE OF INDEPENDENCE

PO5. Demonstrate a high degree of independence, adaptability, and reflective practice in managing continuous professional development aligned with global IT advancements.

PO6. Collaborate effectively with interdisciplinary or multidisciplinary teams in designing, evaluating, and delivering IT solutions.

APPLICATION OF ADVANCED KNOWLEDGE AND SKILLS IN RESEARCH, PROFESSIONAL PRACTICE, OR CREATIVE WORK.

PO7. Translate advanced IT knowledge and research into effective strategies, solutions, technologies, and digital innovations that support organizational goals and societal development.

PO8. Lead and implement IT projects and innovations that align technological, ethical, and societal considerations.

PO9. Apply advanced IT skills in professional and creative work, producing original, innovative, and impactful IT solutions.

PERFORMANCE INDICATORS

The Master of Information Technology (MIT) program at Partido State University is designed to equip students with advanced knowledge and specialized skills in information technology, preparing them for leadership roles in industry, academia, and research. Aligned with CHED Memorandum Order No. 15, s. 2019, the program integrates theoretical foundations with practical applications, ensuring graduates can address complex technological challenges in the rapidly evolving digital landscape.

To assess the effectiveness of the program and measure student competencies, Performance Indicators have been developed for each Learning and Program Outcome. These indicators serve as concrete benchmarks for evaluating students' proficiency in key IT domains, including software development, cybersecurity, data analytics, cloud computing, and IT project management. By meeting these indicators, graduates will demonstrate their ability to innovate, conduct research, lead IT initiatives, and **contribute to the advancement of technology in both organizational and societal contexts.**

Learning Outcomes	Performance Indicators
ADVANCED KNOWLEDGE AND SKILLS IN INFORMATION TECHNOLOGY	
PO1. Apply advanced IT concepts, frameworks, and methodologies to design innovative, secure, and scalable technological solutions that address complex organizational and societal problems.	**a.1** Design and implement enterprise-level IT architectures integrating emerging technologies (AI, IoT, cloud, automation). **a.2** Develop and optimize large-scale databases and data-driven applications supporting analytics and decision-making. **a.3** Apply advanced software engineering practices (microservices, DevOps, CI/CD) to build scalable systems. **a.4** Prototype innovative IT products or services that address societal priorities such as digital transformation, e-governance, smart cities, and sustainability.

	a.5 Apply cybersecurity principles and governance frameworks to protect digital assets and ensure system resilience.
PO2. Demonstrate mastery of specialized IT domains—including systems design, data analytics, cybersecurity, and IT management—through interdisciplinary and industry-aligned approaches.	**b.1** Conduct advanced data analytics using statistical and algorithmic techniques to generate actionable insights. **b.2** Evaluate and implement secure network and cloud infrastructure solutions. **b.3** Create enterprise information systems aligned with strategic organizational requirements. **b.4** Integrate principles from management, computing, and design to build user-centered IT solutions.
DIRECTED RESEARCH	
PO3. Independently conceptualize, design, and conduct high-quality IT research or capstone projects that contribute to innovation, digital transformation, and knowledge creation.	**c.1** Formulate research problems addressing gaps in IT practice or theory. **c.2** Apply quantitative, qualitative, mixed-methods, or design science research approaches aligned with IT standards. **c.3** Produce publishable research output compliant with ethical standards, data protection laws (e.g., Data Privacy Act), and professional guidelines. **c.4** Develop research-based IT solutions validated through experimentation, prototyping, simulation, or usability testing.
PO4. Critically evaluate existing IT literature, technologies, and practices to propose improvements, emerging trends, or novel approaches.	**d.1** Conduct systematic analysis of current IT research to identify trends, limitations, and opportunities. **d.2** Apply critical and analytical thinking to evaluate technological solutions and methodologies.

	d.3 Present scholarly and professional arguments supported by evidence-based reasoning.
LIFELONG LEARNING WITH A HIGH DEGREE OF INDEPENDENCE	
PO5. Demonstrate a high degree of independence, adaptability, and reflective practice in managing continuous professional development aligned with global IT advancements.	**e.1** Obtain IT certifications (AWS, CISSP, Azure, PMP, Google Data Engineer, etc.). **e.2** Engage in industry conferences, research colloquia, hackathons, and professional development programs. **e.3** Maintain a professional portfolio showcasing IT expertise, research output, and innovation projects. **e.4** Demonstrate advanced self-directed learning skills in rapidly evolving IT domains.
PO6. Collaborate effectively with interdisciplinary or multidisciplinary teams in designing, evaluating, and delivering IT solutions.	**f.1** Lead and participate in cross-functional and multicultural IT teams. **f.2** Utilize collaborative development tools (Git, Jira, Miro, Azure DevOps) effectively. **f.3** Demonstrate strong communication, documentation, and leadership skills in team-based IT projects. **f.4** Facilitate knowledge sharing and peer mentoring in IT research or development teams.
APPLICATION OF ADVANCED KNOWLEDGE AND SKILLS IN RESEARCH, PROFESSIONAL PRACTICE, OR CREATIVE WORK	
PO7. Translate advanced IT knowledge and research into effective strategies, solutions, technologies, and digital innovations that support organizational goals and societal development.	**g.1** Produce IT solutions that enhance business processes, efficiency, automation, governance, and public service delivery. **g.2** Develop prototypes aligned with SDGs (e.g., smart environment monitoring, digital governance, sustainable cities). **g.3** Apply project management methodologies (Agile, Scrum, PMBOK) to ensure successful IT project delivery.

PO8. Lead and implement IT projects and innovations that align technological, ethical, and societal considerations.	**h.1** Integrate ethical principles and data governance frameworks into IT designs and decision-making. **h.2** Develop user-centered systems addressing accessibility, inclusiveness, and societal needs. **h.3** Advocate responsible, secure, and sustainable IT practices.
PO9. Apply advanced IT skills in professional and creative work, producing original, innovative, and impactful IT solutions.	**i.1** Create, deploy, and evaluate intelligent systems, digital platforms, or automated solutions that solve real-world problems. **i.2** Prepare technical documentation, research manuscripts, and professional reports meeting global academic and industry standards. **i.3** Present and defend IT solutions to technical panels, stakeholders, and industry partners.

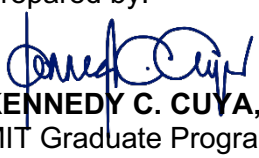
Curriculum Map

Program Outcomes (POs): PO1 Advanced IT solutions; PO2 Specialized domain mastery; PO3 Self-directed research; PO4 Critical evaluation; PO5 Lifelong learning; PO6 Teamwork/leadership; PO7 Organizational/societal application; PO8 Ethics/security/accessibility; PO9 Professional/creative IT work.

Course Code	Course Title	Units	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
MITB01	Object-Oriented Programming	3	**I**	**I**							
MITB02	Data Structure	3	**I**	**I**		**I**					
MITB03	Trends in IT	3		**I**			**I**		**I**		
MIT101	Advanced Database Systems	3	**I/R**	**R**		**I**				**I**	**R**
MIT102	Advanced Operating Systems & Networking	3	**R**	**R**					**I**	**R**	**I**
MIT103	Advanced System Design & Implementation	3	**R**	**R**		**I**		**R**	**I**	**I**	**R**
MIT104	IT Research Methods (Methods of Research)	3			**I/R**	**R**				**R**	
MIT201	Technology & Project Management	3	**I**	**I**			**R**	**R**	**R**		
MIT202	Human-Computer Interaction	3	**I**	**R**		**I**			**I**	**R**	**R**
MIT203	ICT Information Security Administration	3	**R**	**R**					**I**	**M**	**I**
MIT204	Management Information Systems	3	**I**	**R**		**R**			**R**	**I**	
MIT205	Web Engineering	3	**R**	**R**				**I**	**R**	**I**	**R**
MIT301	Introduction to Data Analytics	3	**I**	**R**					**R**		**I**
MIT401	Graduate Project 1	3	**R**	**R**	**R**	**R**	**I**	**R**	**R**	**R**	**R**
MIT402	Graduate Project 2	3	**M**	**M**	**M**	**M**	**R**	**M**	**M**	**M**	**M**

Legend: I = Introduced, R = Reinforced/Developed, M = Mastery/Assessed

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